Ken Yamashiro

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Education

University of Southern California – B.S. Mechanical Engineering, GPA 3.71 — Expected Graduation May 2027

Experience

USC Viterbi School of Engineering — Unmanned Aerial Systems Project — December 2025 – Present
Undergraduate Assistant and Lead Student Engineer — Los Angeles, CA

- Assist Dr.Penkova in characterizing flight dynamics of organic, bio-inspired aerodynamic designs on low-Reynolds-number autonomous VTOL drones in hybrid water/air applications
- Develop electrical systems, program flight dynamics, and essential systems for VTOL flight

Niagara Bottling — May 2026 – Present
Packaging Research and Development Intern — Diamond Bar, CA

- Apply AI and machine learning algorithms to ANSYS FEA tools to simulate and validate lightweighting designs, reducing simulation turnaround time by over 1000 hours for the engineering team
- Integrate simulation outputs into design iterations to support production and environmental improvements

NASA Columbia Scientific Balloon Facility — May 2025 – July 2025
Intern/Undergraduate Assistant — Palestine, TX

- Supported the final production and payload integration of the Payload for Ultrahigh Energy Observations (PUEO) research project, a NASA Pioneers project successfully launched in December 2025
- Performed surface preparation, resolved tolerance conformities, and flight preparation of 300+ components
- Effectively collaborated with over 50 researchers from 11 universities across the U.S., UK, and Taiwan

UH Department of Physics and Astronomy, High Energy Physics Group (HEPG) — September 2023 – August 2025
Lab Tech II and Researcher — Honolulu, HI

- Utilized Siemens SolidEdge, FEA, and drafting to design, fabricate, and test 8+ physics research projects focused on particle physics, radio frequency (RF), and space environments.
- Used CNC, CAM, manual milling, lathes, and additive manufacturing to fabricate 700+ flight hardware parts
- Prepared technical materials and assisted in proposal presentations to NASA, JPL, and various universities

Extracurricular Activities

USC Formula SAE Racing | Powertrain Engineer — August 2025 - Present

- Designed, manufactured, and tested powertrain and cooling subsystem components for the SCR26 race car (600cc inline-4 Yamaha R6) using Siemens NX, ANSYS, and MATLAB
- Selected from 60+ members to present in the design competition at the 2026 Michigan FSAE competition
- Supported assembly, physical testing, and analysis of the SCR26 car, helping place 7th out of 110 teams

Project Experience (More listed on GitHub Website)

Payload for Ultrahigh Energy Observations (PUEO) — NASA

- Antarctic detector for Neutrinos and ultrahigh energy particles, part of the NASA Pioneers Program
- Fabricated 400+ flight components from various materials, programmed CNC toolpaths, and performed FEA
- Drafted part documentation and 100+ reference drawings used during assembly

Askaryian Calorimeter Experiment (ACE) — Fermilab/UH HEPG

- Designed and fabricated an RF testing assembly, resolving part tolerances, assembly, and manufacturability
- Performed FEA on RF assembly to validate waveform compliance and structural stability

Skills and Certifications

Skills: SolidWorks, Siemens NX, Siemens SolidEdge, ANSYS, FEA analysis, Manual/CNC Machining, 3D printing, MATLAB
Experience in: Additive/Subtractive manufacturing, 6061,7075 Al, Brass 360, CF tubes/honeycomb, 100/101 Cu, more
SOLIDWORKS CAD Design Associate Certification, Certificate ID: C-SY4HKFETCM